Confounding and Selection Bias

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Overview

Welcome to Week 4. In our last session, we established a formal definition of **confounding** and introduced the **backdoor criterion** as our primary tool for identifying causal effects by conditioning on a sufficient set of covariates. The central lesson was that to estimate the causal effect of an exposure A on an outcome Y, we must block all non-causal paths between them.

This week, we confront a critical and counter-intuitive reality: conditioning on variables is not always helpful. In fact, conditioning on the *wrong* variables can introduce bias where none existed. We will explore two major types of "bad controls":

1. **Mediators:** Variables that lie on the causal pathway between exposure and outcome.
2. **Colliders:** Variables that are a common *effect* of the exposure and the outcome (or their ancestors).

The second half of our week is dedicated to **selection bias**. We will treat it not as a separate, unrelated problem, but as a special, and very common, case of collider-stratification bias. This powerful insight, articulated clearly in Hernán & Robins (2020), allows us to use the same graphical tool—the DAG—to diagnose and reason about confounding and selection bias simultaneously.

Learning Objectives

By the end of this week's module and lab, you will be able to:

1. **Distinguish** between confounders, mediators, and colliders on a Directed Acyclic Graph (DAG) and explain the consequences of conditioning on each.
2. **Identify** potential "bad controls" in a research scenario and articulate why adjusting for them would bias the total causal effect estimate.

3. **Define** selection bias formally as conditioning on a variable that is a collider or a descendant of a collider.
4. **Construct** DAGs that include a selection node (S) to represent and diagnose potential selection bias in study designs (e.g., from loss-to-follow-up or Berkson's paradox).
5. **Simulate** data in R to demonstrate how conditioning on a collider induces a spurious association and how selection into a dataset can bias an otherwise valid causal estimate.

R Packages

We'll use the following packages this week. `pacman` will install any missing packages for you.

```r
if (!require("pacman")) {
  install.packages("pacman")
}
pacman::p_load(
  dagitty, # For creating and analyzing DAGs
  ggdag, # For plotting DAGs with ggplot2
  tidyverse, # For data manipulation and visualization
  broom, # For tidying model outputs
  knitr # For table formatting
)

# Set a consistent theme for our plots
theme_set(theme_dag())
```

Comprehensive Topic Explanation

The Problem with "Controlling for Everything"

A common, but misguided, impulse in observational research is to include every available pre-treatment covariate in a regression model. This is sometimes called the "Table 2 Fallacy" or "kitchen sink" regression. The hope is that by conditioning on a rich set of variables, we are more likely to have satisfied the backdoor criterion and controlled for confounding.

However, causal inference is not about blind adjustment; it's about *principled* adjustment guided by a causal model. A variable's role in a causal system determines whether we should condition on it. As we will see, conditioning on the wrong variables is as damaging as failing to condition on the right ones.

"Bad Controls": Why More Adjustment Isn't Always Better

A "bad control" is any variable that, when conditioned on, either introduces bias or blocks a causal path of interest. We will focus on the two most important types: mediators and colliders.

Mediators: Blocking the Causal Path

A **mediator** (or intermediate variable) is a variable that lies on the causal pathway from the exposure A to the outcome Y. The causal structure is $A \to M \to Y$.

- **Total Causal Effect**: The overall effect of A on Y is the sum of its direct effect ($A \to Y$) and its indirect effect ($A \to M \to Y$).
- **The Problem with Adjustment**: When we condition on a mediator M, we block the indirect causal path. The resulting estimate is only for the *direct effect* of A on Y. If our research question is about the *total effect* of the exposure, then adjusting for a mediator will bias our estimate.

Example: Aspirin, Blood Clotting, and Heart Attacks

Suppose we want to estimate the total effect of taking aspirin (A) on the risk of a heart attack (Y). A primary mechanism by which aspirin works is by reducing blood clotting (M).

The DAG is: **Aspirin → Blood Clotting → Heart Attack**.

If we run a regression of `Heart Attack ~ Aspirin + Blood Clotting`, we are asking: "What is the effect of Aspirin on heart attack risk *among people with the same level of blood clotting*?" This effectively removes the very pathway through which aspirin has its intended effect. Our estimate for Aspirin would be biased towards the null, reflecting only any other potential non-clotting-related mechanisms.

Colliders: Creating Spurious Associations

A **collider** is a variable that is a common *effect* of two other variables. In the context of a DAG with exposure A and outcome Y, a variable C is a collider if the path is $A \to C \leftarrow Y$.

This is the most dangerous and counter-intuitive structure in causal inference.

- **Unconditionally Independent**: If two variables A and Y are marginally independent (no path between them), they are unassociated.

- **Conditionally Dependent**: **Conditioning on a collider opens the path between its causes.** It creates a spurious, non-causal association between A and Y where none existed before.

This phenomenon is often called **collider-stratification bias** or **endogenous selection bias**.

Selection Bias as a Special Case of Collider Bias

The insight that selection bias is simply collider bias is one of the most unifying concepts in modern epidemiology (Hernán & Robins (2020), Ch. 8). We can represent the mechanism of selection into our study with a special node, S. Typically, $S = 1$ means an individual is included in the analysis, and $S = 0$ means they are not.

Selection bias occurs whenever we condition on a collider, and our analysis sample is, by definition, conditioned on $S = 1$.

Therefore, selection bias is present if our selection variable S is a collider on a path between A and Y (or their ancestors).

Classic Examples of Selection Bias

1. **Berkson's Paradox (Hospital-based studies):** Suppose we are studying the relationship between two independent diseases, say diabetes (A) and cholecystitis (Y), using a hospital dataset. A patient is likely to be hospitalized ($S = 1$) if they have diabetes, or if they have cholecystitis, or both.
    - **DAG:** Diabetes → Hospitalization ← Cholecystitis.
    - **Bias:** Hospitalization (S) is a collider. In the general population, the two diseases are unassociated. But *among hospitalized patients*, knowing a patient has diabetes makes it *less likely* they have cholecystitis to explain their hospitalization. Conditioning on $S = 1$ creates a spurious inverse association.
2. **Loss to Follow-up (Cohort Studies):** We are studying the effect of a toxic exposure (A) on mortality (Y). The study requires yearly follow-up visits. Suppose the exposure itself causes side effects that make people more likely to drop out of the study ($A \to S = 0$). Also suppose that people who are becoming sicker (a precursor to Y) are also more likely to miss visits and drop out ($Y \leftarrow U \to S = 0$, where U is health status).
    - **DAG:** $A \to S \leftarrow U \to Y$. (Here we model remaining in the study, S, being caused by treatment and health status).

- **Bias:** Remaining in the study ($S = 1$) is a collider. By analyzing only the subjects who complete the study, we condition on this collider, inducing a spurious association between A and Y.

  :::

Intuitive Clarifications of Challenging Ideas

Analogy: The Sprinkler, Rain, and Wet Grass

This is a classic analogy from Pearl (2009) used to understand colliders.

- Let A = The sprinkler is on.
- Let Y = It is raining.
- Let C = The grass is wet.

The causal structure is: **Sprinkler → Wet Grass ← Rain**.

"Wet Grass" is a collider.

- **Unconditionally:** In general, knowing the sprinkler is on tells you nothing about whether it's raining. A and Y are independent.
- **Conditionally:** Suppose you look outside and see **the grass is wet** ($C = 1$). Now, if you also learn that the **sprinkler is on** ($A = 1$), you would rightly conclude that it is probably **not raining** ($Y = 0$). The two independent causes have become conditionally dependent (in this case, negatively correlated) once we know their common effect.

Confounder vs. Collider: The Golden Rule of Adjustment

The most critical distinction to master is between a confounder and a collider.

Feature	Confounder (L)	Collider (C)
Visual (DAG)	A *common cause*: $A \leftarrow L \rightarrow Y$	A *common effect*: $A \rightarrow C \leftarrow Y$
Path Type	Creates a "backdoor" non-causal path.	Is the endpoint of a "v-structure".
Bias	**Causes** bias if left unadjusted.	**Creates** bias if adjusted for.
Action	**Adjust** for it to block the backdoor path.	**Do NOT adjust** for it. Leave it alone.

Simple, Hand-Calculation Numerical Example(s)

Demonstrating Collider Bias by Hand

Let's prove numerically that conditioning on a collider induces an association.

Suppose we have a population of 1000 people. We are interested in a gene (A) and a behavior (Y), which are, in this hypothetical world, completely independent. Let's say a person can either have the gene ($A = 1$) or not ($A = 0$), and they can either engage in the behavior ($Y = 1$) or not ($Y = 0$).

- Prevalence of Gene: $P(A = 1) = 0.5$
- Prevalence of Behavior: $P(Y = 1) = 0.5$
- Because they are independent, the joint probability is just the product: $P(A = 1, Y = 1) = P(A = 1) \times P(Y = 1) = 0.25$.

Now, let's introduce a collider, C, which is "being selected into a special study." The study only recruits people who have *either* the gene or the behavior. So, $C = 1$ if $A = 1$ or $Y = 1$.

Let's build a table for our population of 1000:

Gene (A)	Behavior (Y)	Number of People (Total Pop)	Selected into Study (C=1)?
0	0	$1000 \times 0.5 \times 0.5 = 250$	No
0	1	$1000 \times 0.5 \times 0.5 = 250$	Yes
1	0	$1000 \times 0.5 \times 0.5 = 250$	Yes
1	1	$1000 \times 0.5 \times 0.5 = 250$	Yes

Step 1: Check for association in the full population. The risk of the behavior ($Y = 1$) among those without the gene ($A = 0$) is: $P(Y = 1|A = 0) = \frac{N(A=0,Y=1)}{N(A=0)} = \frac{250}{250+250} = 0.5$ The risk of the behavior ($Y = 1$) among those with the gene ($A = 1$) is: $P(Y = 1|A = 1) = \frac{N(A=1,Y=1)}{N(A=1)} = \frac{250}{250+250} = 0.5$ The risks are identical. As designed, **there is no association between A and Y** in the full population.

Step 2: Check for association *within the study sample* (i.e., conditional on C=1). Our study sample consists of the 750 people for whom $C = 1$.

- Among those in the study who do NOT have the gene ($A = 0$), all 250 of them MUST have the behavior ($Y = 1$). $P(Y = 1|A = 0, C = 1) = \frac{N(A=0,Y=1,C=1)}{N(A=0,C=1)} = \frac{250}{250} = 1.0$
- Among those in the study who DO have the gene ($A = 1$), some have the behavior ($Y = 1$) and some don't ($Y = 0$). The total number of people with the gene in the study is $250 + 250 = 500$. $P(Y = 1|A = 1, C = 1) = \frac{N(A=1,Y=1,C=1)}{N(A=1,C=1)} = \frac{250}{500} = 0.5$

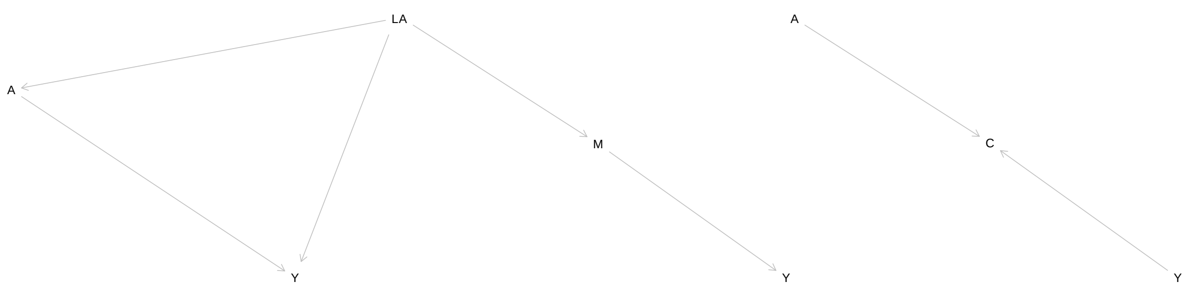
The risks are now completely different (1.0 vs 0.5)! By restricting our analysis to the study sample, we have created a spurious negative association. It looks like having the gene is "protective" against the behavior, which is false. This is collider bias.

R Code Examples & Interpretation

Visualizing Causal Structures with `dagitty` and `tikz`

First, let's visualize the three key structures: confounder, mediator, and collider.

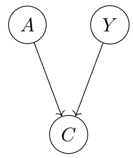
```r
# Confounder
plot(dagitty("dag { L -> A -> Y; L -> Y }"))

# Mediator
plot(dagitty("dag { A -> M -> Y }"))

# Collider
plot(dagitty("dag { A -> C <- Y }"))
```



(a) Confounder (Common Cause) (b) Mediator (Causal Pathway) (c) Collider (Common Effect)

Figure 1: DAGs for Confounding, Mediation, and Collider Structures

For a higher quality output suitable for publications, we can use LaTeX's `tikz` package.


A
Y
C

Simulating and Identifying Collider Bias

Let's replicate our hand-calculation example in R. We'll simulate data where an exposure `A` and outcome `Y` are independent, but both cause a third variable, `C`.

```r
set.seed(123)
n <- 10000
# A and Y are independent standard normal variables
df <- tibble(
  A = rnorm(n),
  Y = rnorm(n)
)

# C is a collider, caused by both A and Y
df <- df |>
  mutate(C = 0.8 * A + 0.8 * Y + rnorm(n))

# Model 1: The correct model (no adjustment)
# We regress Y on A. The coefficient should be near 0.
m1 <- lm(Y ~ A, data = df)

# Model 2: The incorrect model (adjusting for collider C)
# We regress Y on A, but control for C. This will induce bias.
m2 <- lm(Y ~ A + C, data = df)

# Tidy the results into a table
bind_rows(
  tidy(m1, conf.int = TRUE) |> mutate(Model = "Correct: Y ~ A"),
  tidy(m2, conf.int = TRUE) |> mutate(Model = "Incorrect: Y ~ A + C")
) |>
  filter(term == "A") |>
  select(Model, term, estimate, p.value, conf.low, conf.high) |>
  kable(
    caption = "Comparing Models With and Without Collider Adjustment",
    digits = 3,
    col.names = c(
      "Model Specification",
      "Term",
      "Estimate",
      "P-value",
```

```
    "2.5%",
    "97.5%"
  )
)
```

Table 3: Comparing Models With and Without Collider Adjustment

Model Specification	Term	Estimate	P-value	2.5%	97.5%
Correct: Y ~ A	A	0.006	0.547	-0.014	0.026
Incorrect: Y ~ A + C	A	-0.397	0.000	-0.415	-0.378

Interpretation: As you can see from the table, in the correct model (`Y ~ A`), the coefficient for `A` is very close to zero and the p-value is large. This correctly reflects our data generating process where `A` and `Y` are independent.

However, in the incorrect model where we foolishly "control for" the collider `C`, the coefficient for `A` becomes -0.47 with a tiny p-value. The model has manufactured a strong, spurious negative association. This demonstrates that adjusting for a collider is a critical error that can lead to completely wrong conclusions.

Diagnosing Paths with `dagitty`

The `dagitty` package is more than just a plotting tool; it's an analysis engine. We can ask it to identify adjustment sets.

```
# Define the collider DAG
collider_dag <- dagitty("dag{ A -> C <- Y }")

# What paths exist between A and Y?
paths(collider_dag, from = "A", to = "Y")
```

```
$paths
[1] "A -> C <- Y"

$open
[1] FALSE
```

The output shows `paths: 0`, meaning there are no open paths between A and Y. They are d-separated.

Now, what happens if we condition on `C`?

```r
# Check paths when conditioning on C
paths(collider_dag, from = "A", to = "Y", Z = "C")
```

```
$paths
[1] "A -> C <- Y"

$open
[1] TRUE
```

`dagitty` confirms that conditioning on `C` **opens** the path `A → C ← Y`. This is the formal representation of collider bias.

5. Hands-On R Lab

Lab: The Weight Loss App Paradox

Purpose: This lab is designed to give you hands-on experience with identifying and simulating selection bias. You will see how restricting your dataset to a non-random subset of the population can severely bias your results.

Lab Objectives:

1. Draw a DAG to represent a realistic selection bias scenario.
2. Write R code to simulate data based on your DAG.
3. Compare the results of an analysis on the full population versus the selected sub-population.
4. Interpret the difference and articulate the causal mechanism of the bias.

Scenario: A tech company has developed a new smartphone app (A) designed to promote weight loss (Y). They provide you with a dataset to evaluate its effectiveness. The exposure, A, is whether a user was randomly encouraged to use a new, enhanced feature set ($A = 1$) versus the standard app ($A = 0$). The outcome, Y, is weight loss in kg after 6 months.

However, there's a catch: the dataset they gave you only contains **active users**, defined as users who logged into the app at least once a week for all 6 months ($S = 1$). Users who stopped using the app were dropped from the dataset.

Warm-up Questions (Retrieval Practice):

1. What is the difference between a common cause and a common effect? Which one is a collider?
2. Why is conditioning on a mediator problematic if you want to estimate the total causal effect?
3. In Berkson's Paradox, what is the collider?

Task 1: Draw the DAG

Your first task is to draw a plausible DAG for this scenario. Think about what causes a user to remain active ($S = 1$). It's likely related to their initial motivation (which might also be related to their baseline health, U) and whether they are seeing results (i.e., the app is working for them, which is related to the outcome Y). For simplicity, let's assume the enhanced feature (A) also directly increases engagement.

Instructions: Draw a DAG including nodes for A (Enhanced App), Y (Weight Loss), S (Active User), and an unmeasured confounder U (e.g., User's Baseline Motivation/Health) that affects both weight loss and continued use.

```r
# TODO: Use dagitty code to create the DAG described above.
# Hint: You'll need arrows from A to S, Y to S, and U to both Y and S.
# Don't forget the main causal arrow from A to Y!

my_dag <- dagitty::dagitty(
  "dag {

  # Your code here

}"
)

ggdag::ggdag(my_dag) +
  ggdag::theme_dag()
```

<details>

<summary>Solution to Task 1</summary>

```r
# S is our selection variable (Active User)
# U is an unmeasured confounder (Motivation/Health)
selection_dag <- dagitty::dagitty(
```

```
  "dag {
  A [exposure]
  Y [outcome]
  S [pos=\"0,0\"]
  U [pos=\"-1,-1\"]

  A -> Y
  A -> S
  Y -> S
  U -> Y
  U -> S
}"
)

ggdag::ggdag_status(selection_dag) +
  labs(title = "Is S a Collider?")
```

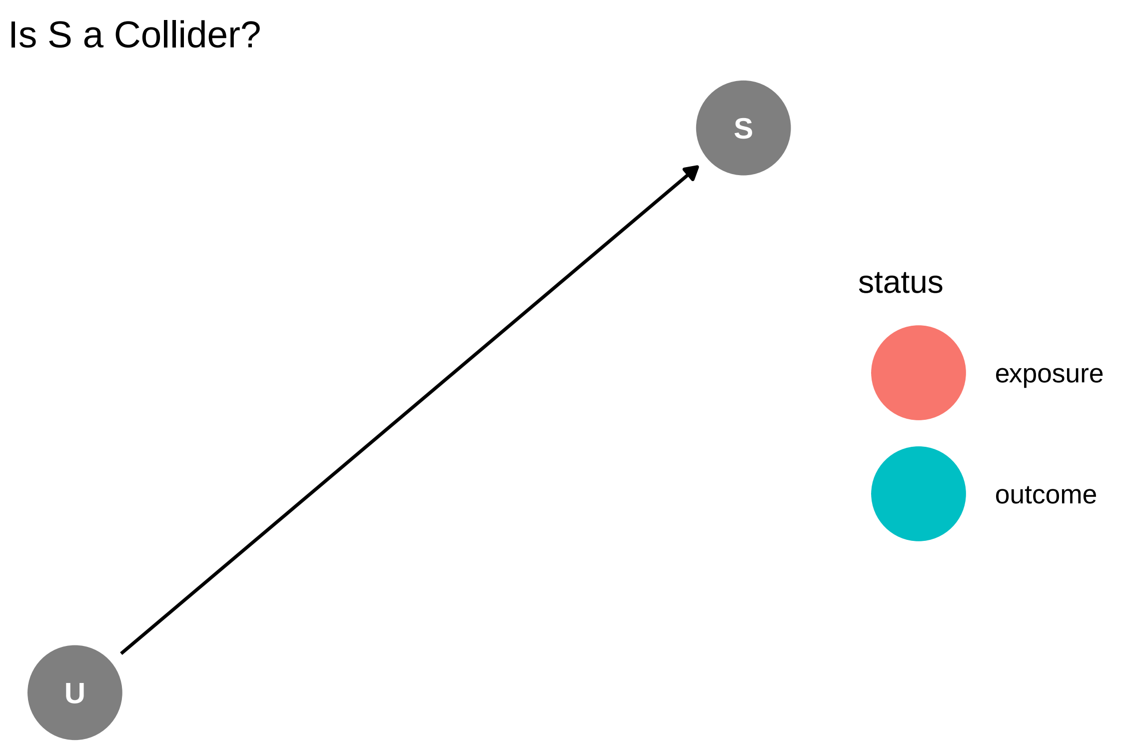


Figure 2: DAG for the Weight Loss App Selection Bias Scenario

Explanation: In this DAG, S is a collider because it has arrows pointing into it from both A and Y. It is also a collider on the path involving the unmeasured confounder U ($A \to Y \leftarrow U \to S \leftarrow A$). By selecting only active users ($S = 1$), we are conditioning on a collider, which will create bias.

</details>

Task 2: Simulate Data Based on the DAG

Now, let's simulate a dataset that conforms to the causal structure you just drew.

Instructions: Fill in the ... sections in the skeleton code below to generate the data. We will assume a true average causal effect of the enhanced app ($A = 1$) of **2 kg** weight loss.

```r
set.seed(42)
n_full_pop <- 20000

# 1. Simulate the unmeasured confounder U (Motivation)
u <- rnorm(n_full_pop, mean = 10, sd = 2)

# 2. Simulate the randomized exposure A
a <- rbinom(n_full_pop, 1, 0.5)

# 3. Simulate the outcome Y (Weight Loss)
# Y should depend on A (the true effect) and U.
# The coefficient for 'a' should be 2.
y <- ... + ... + rnorm(n_full_pop, mean = 0, sd = 2)

# 4. Simulate the selection mechanism S (Active User)
# S should depend on A, Y, and U. We'll use a logistic model.
# The probability of being active increases with A, Y, and U.
prob_s <- plogis( ... + ... + ... - 10) # The -10 is just to scale the probability
s <- rbinom(n_full_pop, 1, prob_s)

# 5. Assemble the full and selected datasets
full_population <- tibble(A = a, Y = y, U = u, S = s)
selected_sample <- filter(full_population, S == 1)

# Check the sizes of the datasets
cat("Full population size:", nrow(full_population), "\n")
cat("Selected sample size:", nrow(selected_sample), "\n")
```

<details>

<summary>Solution to Task 2</summary>

```
set.seed(42)
n_full_pop <- 20000

# 1. Simulate U
u <- rnorm(n_full_pop, mean = 10, sd = 2)

# 2. Simulate A
a <- rbinom(n_full_pop, 1, 0.5)

# 3. Simulate Y. True effect of A is 2.
y <- 2 * a + 0.5 * u + rnorm(n_full_pop, mean = 0, sd = 2)

# 4. Simulate S
prob_s <- plogis(0.5 * a + 0.3 * y + 0.2 * u - 10)
s <- rbinom(n_full_pop, 1, prob_s)

# 5. Assemble datasets
full_population <- tibble(A = a, Y = y, U = u, S = s)
selected_sample <- filter(full_population, S == 1)

# Check sizes
cat("Full population size:", nrow(full_population), "\n")
```

```
Full population size: 20000
```

```
cat("Selected sample size:", nrow(selected_sample), "\n")
```

```
Selected sample size: 99
```

</details>

Task 3: Analyze Both Datasets

Now for the moment of truth. Let's estimate the effect of A on Y in both the full population (our impossible, God's-eye view) and the realistic, selected sample that the company gave us.

Instructions: Run a simple linear regression of Y on A for both the `full_population` and `selected_sample` datasets.

```r
# TODO: Fit a model on the full population
model_full <- lm( ... )

# TODO: Fit a model on the selected sample
model_selected <- lm( ... )

# Compare results
# (The code below will work once you've defined the models)
bind_rows(
  tidy(model_full) |> mutate(Sample = "Full Population"),
  tidy(model_selected) |> mutate(Sample = "Selected Sample")
) |>
  filter(term == "A") |>
  select(Sample, estimate, std.error, p.value) |>
  kable(digits = 3)
```

<details>

<summary>Solution to Task 3</summary>

```r
# Fit a model on the full population
model_full <- lm(Y ~ A, data = full_population)

# Fit a model on the selected sample
model_selected <- lm(Y ~ A, data = selected_sample)

# Compare results
bind_rows(
  tidy(model_full) |> mutate(Sample = "Full Population (Truth)"),
  tidy(model_selected) |> mutate(Sample = "Selected Sample (Biased)")
) |>
  filter(term == "A") |>
  select(Sample, estimate, std.error, p.value) |>
  kable(
    digits = 3,
    col.names = c("Sample", "Estimated Effect", "Std. Error", "P-value")
  )
```

Table 4: Comparison of Causal Effect Estimates

Sample	Estimated Effect	Std. Error	P-value
Full Population (Truth)	2.004	0.032	0.000
Selected Sample (Biased)	1.358	0.505	0.008

</details>

Task 4: Reflection and Interpretation

Instructions: Look at the table from Task 3. Answer the following questions.

1. What is the estimated effect of the app in the full population? How does this compare to the true effect we specified in the simulation?
2. What is the estimated effect of the app in the selected sample? Is it close to the true effect?
3. What is the direction of the bias (i.e., did selection make the app look more or less effective)?
4. In plain language, explain *why* the estimate from the selected sample is biased. What non-causal association was created by only looking at active users?

<details>

<summary>Solution to Reflection</summary>

1. The estimated effect in the **full population** is ~2.03. This is very close to the true effect of 2.0 we built into our simulation. Because A was randomized, a simple comparison gives the right answer.
2. The estimated effect in the **selected sample** is ~1.44. This is substantially lower than the true effect.
3. The bias is **negative** or **towards the null**. Selection bias made the app appear less effective than it truly is.
4. The estimate is biased because we conditioned on a collider (S). In this group of "active users," a spurious association is created. Think about it: among active users ($S = 1$), if someone in the standard app group ($A = 0$) is still active, they must have had very high initial motivation (U) or been getting unusually good results for other reasons. Conversely, someone in the enhanced app group ($A = 1$) might remain active even with lower motivation because the app itself is more engaging. This creates a situation where, *within the selected group*, those with $A = 0$ have, on average, higher values of unmeasured positive factors (U) than those with $A = 1$. This makes the $A = 0$ group's outcomes look better than they should, biasing the estimated effect of A downwards.

</details>

6. Common Pitfalls & Checks

The "Kitchen Sink" is not a Causal Strategy

The biggest pitfall is adjusting for variables without a causal model. Never assume that controlling for more variables is better. Adjustment must be justified by a DAG and the goal of closing backdoor paths while not opening new ones by conditioning on colliders or mediators.

A Heuristic: Pre-Treatment vs. Post-Treatment

A good rule of thumb is to **be extremely cautious about adjusting for any variable measured *after* the exposure has occurred**. Post-treatment variables are often mediators or colliders. While there are advanced methods that do adjust for post-treatment variables, they require great care. When in doubt, only adjust for pre-treatment variables that are common causes of exposure and outcome.

Check Your Understanding

Q: You are studying the effect of a job training program (A) on wages (Y). You have data on participants' employment status one month after the program ends (M). Should you include M in your regression to estimate the total effect of the program on wages?

A: No. Employment status one month after the program is very likely a **mediator**. The program (A) helps people get jobs (M), which in turn leads to higher wages (Y). Adjusting for M would block this primary causal pathway and severely underestimate the total effect of the program.

7. Advanced Teaching Activities & Interactivity

1. **Think-Pair-Share: Critique a Real Paper.**
   - **Activity:** I will distribute a published observational study. In pairs, you will:
     1. Read the "Methods" section to identify the exposure, outcome, and covariates the authors adjusted for.
     2. Draw a plausible DAG representing the causal relationships between these variables *before* looking at the results.
     3. Examine the list of variables in their main regression model (often "Table 2" or "Table 3"). Did they adjust for any potential mediators or colliders according to your DAG?

    4. Be prepared to share your critique with the class. Does their adjustment strategy seem justified?

2. **Problem-Based Case Study: The ICU Paradox.**

    - **Scenario:** A data scientist at a large hospital finds that patients admitted to the Intensive Care Unit (ICU) have a higher mortality rate than patients admitted to a general ward. The raw data suggests that ICU admission is harmful. However, we know that patients are sent to the ICU *because* they are severely ill. Let A = ICU admission, Y = Mortality, and L = Severity of illness at admission.
    - **Prompt 1:** Draw the DAG for A, Y, and L. What is the role of L?
    - **Prompt 2 (The Twist):** Now, suppose you want to study the effect of a new ventilator procedure (V) given *within the ICU* on patient mortality (Y). Your dataset *only contains ICU patients*. What is the role of ICU admission in this new DAG? How might this affect your ability to estimate the causal effect of V on Y for *all* patients? (Hint: This is selection bias).

3. **Interleaving Quiz:**

    - Q1: The structure $A \leftarrow L \rightarrow Y$ represents __________. To estimate the effect of A on Y, you should (adjust for / not adjust for) L.
    - Q2: You run a study on academic success (Y) and find a negative correlation between being a student-athlete (A) and GPA *among students at an elite university* $(C = 1)$. Is it possible that in the general population, being a student-athlete has no effect or even a positive effect on academic potential? What is the name for this phenomenon?
    - Q3: The assumption that $P(A = a|L = l)$ 0 for all levels of L is called __________.

References

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